Keyboard switch replacement

Replace a faulty E-key switch on a compatible hot-swap keyboard.

DOCUMENT	PERFORMED BY	RELEASED BY
SOP-KBD-01 / v1.1	Service technician	Service supervisor

Scope & prerequisites

For trained technicians using a verified hot-swap mechanical board and the manufacturer's model-specific instructions. Soldering, battery work and case disassembly are outside this procedure. Confirm switch compatibility before service. This is a TARA demonstration procedure, pending approval before operational use.

PREPARE THE DEVICE

1. Identify the device

Read the keyboard model from its underside label. Match the model to the work order and record the affected key as E.

Checkpoint: Keyboard model and target key match the service record.

2. Prepare the work area

Disconnect the USB cable and switch off wireless power. Place the keyboard on a clean, dry, ESD-safe work surface. The technician must confirm the device is powered off before using either puller.

Checkpoint: Cable disconnected, power off, and work surface ready.

3. Check tools and replacement parts

Check that a keycap puller, switch puller, parts tray and compatible replacement switch are present. Compare the replacement type and pin configuration with the model specification.

Checkpoint: Required tools are available and the replacement is compatible.

4. Remove the keycap

Position the keycap puller around the E keycap and lift it straight upward with controlled force. Place the keycap in the parts tray. Stop if resistance is unusual or any part begins to deform.

Checkpoint: The E keycap is removed and retained without visible damage.

Replace, verify & release

Continue only after the preparation checkpoints are complete.

5. Inspect the exposed switch

Inspect the exposed switch housing and surrounding plate for visible damage, debris or a displaced housing. Capture a clear view of the complete switch. If the view is obscured, adjust the lighting and inspect again before continuing.

Checkpoint: The housing and surrounding area are clearly visible and assessed.

6. Remove and install the switch

Use the switch puller to release the retaining clips and lift the faulty switch straight out. Check the replacement pins are straight, align them with the socket, and seat the switch gently. Never force it. Inspect that it sits flush, then align and refit the E keycap. Stop and escalate any damaged socket or unusual resistance.

Checkpoint: Replacement seated evenly; keycap refitted and moving freely.

7. Record the functional test

Reconnect the keyboard and open a key tester. Press and fully release E ten times. Record the number of presses and registrations, and attach the test result to the work order. Pass only when every press registers exactly once and the key releases normally. A failed test requires further diagnosis before release.

Checkpoint: 10 presses, 10 single registrations, and a saved test result.

8. Review and approve release

A supervisor must review the device identity, replacement details, inspection and functional test. Record the approver's name and completion date. Approve return to the customer only after all checkpoints pass; otherwise keep the work order open and record the exception.

Checkpoint: Named approval is recorded, or the open exception is documented.

SERVICE RECORD

Work order: _____ Technician: _____

Test result: PASS / HOLD Evidence reference: _____

Supervisor: _____ Completion date: _____

Hardware reference: Keychron FAQ - switch compatibility and safe replacement. Follow the exact model manual. The ten-press check and release workflow are TARA demo conventions, not manufacturer certification.